

OpenEXR Attributes Specification

OpenTrackIO Source Mapping Extension

Version: 0.5 DRAFT 2 Date: 2026-07-20

Status: FOR REVIEW

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1. General

1.1 Scope

This document defines extensions to the following metadata, based on the [OpenEXR Attributes Base Specification for Camera Acquisition Metadata](#).

- [OpenTrackIO Documentation v1.0.1](#) (<https://ris-pub.smpite.org/ris-osvp-metadata-camdkit/>)

1.2 Source Metadata

This specification primarily covers the following information to be stored in OpenEXR Header Attributes:

- Metadata Sets defined in [OpenTrackIO Documentation v1.0.1](#) specificaion.

This document does not specify the method for obtaining the above metadata; however, it may be captured via a JSON file or over an IP network.

1.3 OpenEXR Header Attributes

1.3.1 OpenEXR Specificatiopn

The OpenEXR Header Attributes defined in this specification SHALL conform to the [OpenEXR Attributes Base Specification for Camera Acquisition Metadata](#), which further conforms to the following specification:

- OpenEXR Specification, version 3.x (<https://openexr.com/en/latest/index.html>)

Equivalent specifications are also defined in the following:

- SMPTE ST 2065-4:2023, ACES Image Container File Layout (<https://pub.smpite.org/doc/st2065-4/20220404-pub/>)
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1.4 Target USD Mapping

This extension participates in the following USD destinations:

- USD Stage Metadata
 - Camera Prim xform
 - [UsdGeom.Camera](#) built-in attributes
 - Camera Prim [customData](#)
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2. Convention Policy

This section describes the policy for defining extensions to OpenEXR attributes.

2.1 Base Policy

The policy of the Base Specification ([OpenEXR Attributes Base Specification for Camera Acquisition Metadata](#)) SHALL be followed.

2.2 Namespace

All OpenEXR attributes SHALL have a namespace identifier beginning with `acq:` . The Base Specification further defines the following second-level sub-namespaces:

- MXF sub-namespace (`mxf:`)
- Lens sub-namespace (`lens:`)
- Camera sub-namespace (`camera:`)
- Imager sub-namespace (`imager:`)
- Color sub-namespace (`color:`)
- Slate sub-namespace (`slate:`)

In addition to above, this specification defines the following second-level sub-namespaces:

- OpenTrackIO sub-namespace (`opentrackio:`)

2.3 Preference for Unified Attributes

For OpenEXR attributes in the sub-namespaces extended by this document, attributes that represent common concepts SHOULD be normalized and stored as vendor-neutral attributes defined by the Base Specification. In such cases, specification-specific or vendor-specific attributes belonging to the sub-namespaces extended by this document MAY also be stored together with the normalized attributes.

If there is a concern that information may be lost by normalization into vendor-neutral attributes, specification-specific or vendor-specific information SHOULD be referenced as necessary.

Furthermore, metadata that can be represented using OpenEXR Standard Attributes SHOULD be expressed using those attributes. In such cases, vendor-neutral attributes defined by this specification and specification/vendor-specific attributes MAY also be stored together with the OpenEXR Standard Attributes.

3 Metadata Mapping

3.1 Overview

The specification-specific attributes defined in this specification use the namespaces shown in the table below.

Namespace Identifier	Description for the source information
<code>acq:opentrackio:</code>	SMPTE OpenTrackIO specification-specific metadata

Additional Sub-namespace Identifiers may be appended after the Namespace Identifier described above, as needed.

3.2 Schema Identifier Attributes

3.2.1 OpenTrackIO Schema Identifier Attributes

The following schema identifier SHALL be used to describe attributes in the OpenTrackIO namespace.

Attribute Name	Type	Requirement	Value
<code>acq:opentrackio:schema:extensionNamestandard</code>	<code>string</code>	REQUIRED	"OpenTrackIO Source Mapping Extension"
<code>acq:opentrackio:schema:extensionVersion</code>	<code>string</code>	REQUIRED	"0.50"

When the source schema is known, the following SHOULD also be present.

Attribute Name	Type	Requirement	Value
<code>acq:opentrackio:payload:schemaName</code>	<code>string</code>	REQUIRED	"OpenTrackIO"

Attribute Name	Type	Requirement	Value
<code>acq:opentrackio:payload:schemaVersion</code>	string	REQUIRED	"1.0.1"
<code>acq:opentrackio:payload:camdkitVersion</code>	string	OPTIONAL	

See also Section 4.

3.3 OpenTrackIO Attributes

3.1.1 OpenTrackIO Native Notation

When storing information obtained from each item of the OpenTrackIO as OpenTrackIO-specific OpenEXR Attributes, the definitions in the table below SHALL be followed.

However, where normalization to vendor-neutral OpenEXR Attributes defined in the Base Specification is possible, they SHOULD be normalized in accordance with Section 3.1.2.

Also, where normalization to OpenEXR Standard Attributes is possible, they SHOULD be normalized in accordance with Section 3.1.3.

OpenTrackIO Property Name	OpenTrackIO OpenEXR Attribute Name	OpenEXR Attribute Type
static:duration	<code>acq:opentrackio:static:duration</code>	rational
static:camera: captureFrameRate	<code>acq:opentrackio:static:camera:captureFrameRate</code>	rational
static:camera: activeSensorPhysicalDimensions	<code>acq:opentrackio:static:camera: activeSensorPhysicalDimensions</code>	v2f
static:camera: activeSensorResolution	<code>acq:opentrackio:static:camera: activeSensorResolution</code>	v2i
static:camera: make	<code>acq:opentrackio:static:camera:make</code>	string
static:camera: model	<code>acq:opentrackio:static:camera:model</code>	string
static:camera: serialNumber	<code>acq:opentrackio:static:camera:serialNumber</code>	string
static:camera: firmwareVersion	<code>acq:opentrackio:static:camera:firmwareVersion</code>	string
static:camera: label	<code>acq:opentrackio:static:camera:label</code>	string
static:camera: anamorphicSqueeze	<code>acq:opentrackio:static:camera:isoSpeed</code>	rational
static:camera: isoSpeed	<code>acq:opentrackio:static:camera:isoSpeed</code>	unsigned int
static:camera: fdllink	<code>acq:opentrackio:static:camera:fdllink</code>	string
static:camera: shutterAngle	<code>acq:opentrackio:static:camera:shutterAngle</code>	float

OpenTrackIO Propetery Name	OpenTrackIO OpenEXR Attribute Name	OpenEXR Attribute Type
static:lens: distortionOverscanMax	acq:opentrackio:static:lens:distortionOverscanMax	float
static:lens: undistortionOverscanMax	acq:opentrackio:static:lens:undistortionOverscanMax	float
static:lens: make	acq:opentrackio:static:lens:make	string
static:lens: model	acq:opentrackio:static:lens:model	string
static:lens: serialNumber	acq:opentrackio:static:lens:serialNumber	string
static:lens: firmwareVersion	acq:opentrackio:static:lens:firmwareVersion	string
static:lens: nominalFocalLength	acq:opentrackio:static:lens:nominalFocalLength	float
static:lens: calibrationHistory	acq:opentrackio:static:lens:calibrationHistory	stringvector
static:tracker: make	acq:opentrackio:static:tracker:make	string
static:tracker: model	acq:opentrackio:static:tracker:model	string
static:tracker: serialNumber	acq:opentrackio:static:tracker:serialNumber	string
static:tracker: firmwareVersion	acq:opentrackio:static:tracker:firmwareVersion	string
static:tracker: notes	acq:opentrackio:tracker:notes	string
static:tracker: recording	acq:opentrackio:tracker:recording	unsigned char
static:tracker: slate	acq:opentrackio:tracker:slate	string
static:tracker: status	acq:opentrackio:tracker:status	string
timing: mode	acq:opentrackio:timing:mode	string
timing:recordedTimestamp: seconds	acq:opentrackio:timing:recordedTimestamp:seconds	unsigned long
timing: recordedTimestamp: nanoseconds	acq:opentrackio:timing:recordedTimestamp:nanoseconds	unsigned int

OpenTrackIO Propetery Name	OpenTrackIO OpenEXR Attribute Name	OpenEXR Attribute Type
timing: sampleRate	acq:opentrackio:timing:sampleRate	rational
timing:sampleTimestamp: seconds	acq:opentrackio:timing:sampleTimestamp:seconds	unsigned long
timing:sampleTimestamp: nanoseconds	acq:opentrackio:timing:sampleTimestamp:nanoseconds	unsigned int
timing: sequenceNumber	acq:opentrackio:timing:sequenceNumber	unsigned int
timing:synchronization: locked	acq:opentrackio:timing:synchronization:locked	unsigned char
timing:synchronization: source	acq:opentrackio:timing:synchronization:source	string
timing:synchronization: frequency	acq:opentrackio:timing:synchronization:frequency	rational
timing:synchronization: offsets:type	acq:opentrackio:timing:synchronization:offsets: type	float
timing:synchronization: offsets:rotation	acq:opentrackio:timing:synchronization:offsets: rotation	float
iming:synchronization: offsets:lensEncoders	acq:opentrackio:timing:synchronization:offsets: lensEncoders	float
timing:synchronization: present	acq:opentrackio:timing:synchronization:present	unsigned char
timing:synchronization: ptp:profile	acq:opentrackio:timing:synchronization:ptp: profile	string
timing:synchronization: ptp:domain	acq:opentrackio:timing:synchronization:ptp: domain	unsigned char
timing:synchronization: ptp:leaderIdentity	acq:opentrackio:timing:synchronization:ptp: leaderIdentity	string
timing:synchronization: ptp:leaderPriorities:priority1	acq:opentrackio:timing:synchronization:ptp: leaderPriorities:priority1	unsigned char
timing:synchronization: ptp:leaderPriorities:priority2	acq:opentrackio:timing:synchronization:ptp: leaderPriorities:priority2	unsigned char
timing:synchronization: ptp:leaderAccuracy	acq:opentrackio:timing:synchronization:ptp: leaderAccuracy	unsigned
timing:synchronization: ptp:leaderTimeSource	acq:opentrackio:timing:synchronization:ptp: leaderTimeSource	string
timing:synchronization: ptp:meanPathDelay	acq:opentrackio:timing:synchronization:ptp: meanPathDelay	float
timing:synchronization: ptp:vlan	acq:opentrackio:timing:synchronization:ptp: vlan	unsigend int

OpenTrackIO Propetery Name	OpenTrackIO OpenEXR Attribute Name	OpenEXR Attribute Type
timing:timecode: hours	acq:opentrackio:timing:timecode:hours	unsigned char
timing:timecode: minutes	acq:opentrackio:timing:timecode:minutes	unsigned char
timing:timecode: seconds	acq:opentrackio:timing:timecode:seconds	unsigned char
timing:timecode: frames	acq:opentrackio:timing:timecode:frames	unsigned char
timing:timecode: frameRate	acq:opentrackio:timing:timecode:frameRate	rational
iming:timecode: subFrame	acq:opentrackio:timing:timecode:subFrame	unsigned int
timing:timecode: dropFrame	acq:opentrackio:timing:timecode:dropFrame	unsigned char
lens:custom	acq:opentrackio:lens:custom	floatVector
lens:distortion: model	acq:opentrackio:lens:distortion:<N>:model	string
lens:distortion: radial	acq:opentrackio:lens:distortion:<N>:radial	floatVector
lens:distortion: tangential	acq:opentrackio:lens:distortion:<N>:tangential	floatVector
lens:distortion: overscan	acq:opentrackio:lens:distortion:<N>:overscan	floatVector
lens:encoders: iris	acq:opentrackio:lens:encoders:iris	float
lens:encoders: focus	acq:opentrackio:lens:encoders:focus	float
lens:encoders: zoom	acq:opentrackio:lens:encoders:zoom	float
lens: entrancePupilOffset	acq:opentrackio:lens:entrancePupilOffset	float
lens: exposureFalloff:a1	acq:opentrackio:lens:exposureFalloff:a1	float
lens: exposureFalloff:a2	acq:opentrackio:lens:exposureFalloff:a2	float
lens: exposureFalloff:a3	acq:opentrackio:lens:exposureFalloff:a3	float
lens: fStop	acq:opentrackio:lens:fStop	float

OpenTrackIO Propetery Name	OpenTrackIO OpenEXR Attribute Name	OpenEXR Attribute Type
lens: pinholeFocalLength	acq:opentrackio:lens:pinholeFocalLength	float
lens: focusDistance	acq:opentrackio:lens:focusDistance	float
lens: projectionOffset	acq:opentrackio:lens:projectionOffset	v2f
lens: rawEncoders:focus	acq:opentrackio:lens:rawEncoders:focus	unsigned ing
lens: rawEncoders:iris	acq:opentrackio:lens:rawEncoders:iris	unsigned ing
lens: rawEncoders:zoom	acq:opentrackio:lens:rawEncoders:zoom	unsigned ing
lens: tStop	acq:opentrackio:lens:tStop	float
protocol: name	acq:opentrackio:protocol:name	string
protocol: version	acq:opentrackio:protocol:version	string
sampleId	acq:opentrackio:sampleId	string
sourceId	acq:opentrackio:sourceId	string
sourceNumbe	acq:opentrackio:sourceNumber	unsigned int
relatedSampleIds	acq:opentrackio:relatedSampleIds	stringvector
globalStage	acq:opentrackio:globalStage	floatVector
transforms	acq:opentrackio:transforms:<N>	M44f
transforms:id	acq:opentrackio:transforms:<N>:id	string
transforms:translation	acq:opentrackio:transforms:<N>:translation	v3f
transforms:rotation	acq:opentrackio:transforms:<N>:rotation	v3f

Note that, in the attribute names for lens:distortion:* properties and transforms properties, <N> shall be a zero-based numeric index that identifies an element of the object array.

3.1.2 Normalization to Vendor-Neutral Camera Attributes

Among the items in the OpenTrackIO, those listed in the table below can be stored as the corresponding vendor-neutral OpenEXR Attributes defined in Base Specification, and therefore these SHOULD take precedence.

OpenTrackIO Propetery Name	Vendor-Neutral OpenEXR Attribute Name	OpenEXR Attribute Type	Conversion
static:camera: captureFrameRate	acq:camera:captureFrameRate	rational	

OpenTrackIO Propetery Name	Vendor-Neutral OpenEXR Attribute Name	OpenEXR Attribute Type	Conversion
static:camera: activeSensorPhysicalDimensions	acq:imager:overallDimension	v2f	
static:camera: activeSensorResolution	acq:imager:effectiveRectangle	box2i	
static:camera: make	acq:camera:make	string	
static:camera: model	acq:camera:model	string	
static:camera: serialNumber	acq:camera:serialNumber	string	
static:camera: firmwareVersion	acq:camera:firmwareVersion	string	
static:camera: label	acq:camera:label	string	
static:camera: isoSpeed	acq:camera:isoSensitivity	unsigned int	
static:camera: shutterAngle	acq:camera:exposureTime	float	
static:lens: make	acq:lens:make	string	
static:lens: model	acq:lens:model	string	
static:lens: serialNumber	acq:lens:serialNumber	string	
static:lens: firmwareVersion	acq:lens:firmwareVersion	string	
static:lens: nominalFocalLength	acq:lens:nominalFocalLength	float	
lens: entrancePupilOffset	acq:lens:entrancePupilOffset	float	
lens: fStop	acq:lens:irisFNumber	float	
lens: pinholeFocalLength	acq:lens:pinholeFocalLength	float	
lens: focusDistance	acq:lens:focusDistance	float	
lens: tStop	acq:lens:irisTStop	float	

3.1.3 Normalization to OpenEXR Standard Attributes

Among the items in the OpenTrackIO, those listed in the table below can be stored as the corresponding OpenEXR Standard Attributes, and therefore these SHOULD take precedence.

OpenTrackIO Propetery Name	OpenEXR Standard Attribute Name	OpenEXR Attribute Type	Conversion
static:camera: captureFrameRate	captureRate	rational	
static:camera: activeSensorResolution	sensorAcquisitionRectangle	box2i	
static:camera: make	cameraMake	string	
static:camera: model	cameraModel	string	
static:camera: serialNumber	cameraSerialNumber	string	
static:camera: firmwareVersion	cameraFirmwareVersion	string	
static:camera: label	cameraLabel	string	
static:camera: isoSpeed	isoSpeed	unsigned int	
static:camera: shutterAngle	expTime	float	
static:lens: make	lensMake	string	
static:lens: model	lensModel	string	
static:lens: serialNumber	lensSerialNumber	string	
static:lens: firmwareVersion	lensFirmwareVersion	string	
static:lens: nominalFocalLength	nominalFocalLength	float	
lens: entrancePupilOffset	entrancePupilOffset	float	
lens: fStop	aperture	float	
lens: pinholeFocalLength	pinholeFocalLength	float	
lens: focusDistance	focus	float	

OpenTrackIO Propetery Name	OpenEXR Standard Attribute Name	OpenEXR Attribute Type	Conversion
lens: projectionOffset	sensorCenterOffsett	v2f	
lens: tStop	tStop	float	
transforms	worldToCamera	M44f	inverse coordinate

4 OpenTrackIO Payload Preservation

As another approach, the complete OpenTrackIO payload MAY be preserved as UTF-8 JSON instead of mapping opentrackio to individual OpenEXR attributes.

Attribute Key	Type	Requirement	Description
opentrackio:payload:json	string	RECOMMENDED	Full source payload JSON
opentrackio:payload:encoding	string	OPTIONAL	"utf-8"
opentrackio:payload:schemaName	string	OPTIONAL	Source schema name
opentrackio:payload:schemaVersion	string	OPTIONAL	Source schema version
opentrackio:payload:camdkitVersion	string	OPTIONAL	camdkit version

For the schema indication, see also Section 3.2.

When both flat `acq:opentrackio:*` attributes and `acq:opentrackio:payload:json` exist:

1. Flat `acq:opentrackio:*` attributes SHALL be used for mapping.
2. `acq:opentrackio:payload:json` SHALL be treated as provenance and round-trip data.
3. Payload JSON MAY be used only when no flat attribute exists.

5. USD Mapping

5.1 USD Mapping Source Selection

USD authoring SHALL use an explicit source selection policy when both `acq:*` and `opentrackio:*` values are available.

5.1.1 Mapping Policy Attributes

The following optional attributes MAY document the selected policy.

Attribute Key	Type	Requirement	Description
opentrackio:mapping:usdCameraIntrinsicsSource	string	OPTIONAL	"acq", "opentrackio", "preferAcq", "preferOpenTrackIO"
opentrackio:mapping:usdCameraTransformSource	string	OPTIONAL	"opentrackio", "acqPosture", "none"
opentrackio:mapping:openexrStandardAttributeSource	string	OPTIONAL	"acq", "opentrackio", "preferAcq", "preferOpenTrackIO"

Recommended default:

```
opentrackio:mapping:usdCameraIntrinsicsSource = "preferAcq"
opentrackio:mapping:usdCameraTransformSource = "opentrackio"
opentrackio:mapping:openexrStandardAttributeSource = "preferAcq"
```

This default means:

- camera body metadata uses `acq:*` when available
- tracking transform uses `opentrackio:*`
- OpenEXR Standard Attributes prefer `acq:*`
- OpenTrackIO values may be used when `acq:*` is absent or explicitly selected

5.2. USD Stage Metadata Mapping

5.2.1 Source Priority for `timeCodesPerSecond`

If OpenTrackIO is selected for timing:

1. `acq:opentrackio:timing:sampleRate`
2. `acq:opentrackio:static:camera:captureFrameRate`
3. OpenEXR Standard Attribute `framesPerSecond`

If Base timing is selected:

1. `acq:mxf:timeline:editRate`
2. `acq:mxf:video:sampleRate`
3. OpenEXR Standard Attribute `framesPerSecond`

5.2.2 Source Priority for `framesPerSecond`

If OpenTrackIO is selected:

1. `acq:opentrackio:timeline:captureFrameRate`
2. `acq:opentrackio:timing:sampleRate`

If `acq:*` is selected:

1. `acq:camera:captureFrameRate`
2. Base Specification fallback rules

5.3 UsdGeom.Camera Mapping

5.3.1 If `acq:*` Is Selected

USD Attribute	Source
<code>focalLength</code>	<code>acq:lens:effectiveFocalLength</code> , then <code>acq:lens:nominalFocalLength</code>
<code>fStop</code>	<code>acq:lens:irisFNumber</code>
<code>focusDistance</code>	<code>acq:lens:focusDistance</code>
<code>horizontalAperture</code> / <code>verticalAperture</code>	<code>acq:imager:effectiveDimension</code>
<code>shutter:open</code> / <code>shutter:close</code>	<code>acq:camera:exposureTime</code>

5.3.2 If `opentrackio:*` Is Selected

USD Attribute	Source
<code>focalLength</code>	<code>acq:opentrackio:static:lens:nominalFocalLength</code>
<code>fStop</code>	<code>acq:opentrackio:lens:fStop</code>
<code>focusDistance</code>	<code>acq:opentrackio:lens:focusDistance</code>
<code>horizontalAperture</code> <code>verticalAperture</code>	<code>acq:opentrackio:static:camera:activeSensorPhysicalDimensions</code>
<code>shutter:open</code> <code>shutter:close</code>	<code>acq:opentrackio:static:camera:shutterAngle</code>

5.3.3 Shutter Mapping

```
shutterDurationFrames = exposureTimeSeconds * framesPerSecond
shutter:open = -shutterDurationFrames / 2
shutter:close = shutterDurationFrames / 2
```

5.4 USD Camera Transform Mapping

OpenTrackIO tracking is the preferred source for USD Camera Prim transform when selected and valid.

Source priority:

- 1. inverse of `acq:opentrackio:transforms`
- 2. `acq:opentrackio:transforms:translation` and `acq:opentrackio:transforms:rotation`

The selected transform SHALL be used only when tracking is valid according to tracker status.

Only the selected transform source attribute SHALL be considered consumed.

Unselected transform candidates MAY be preserved in Camera Prim `customData` .